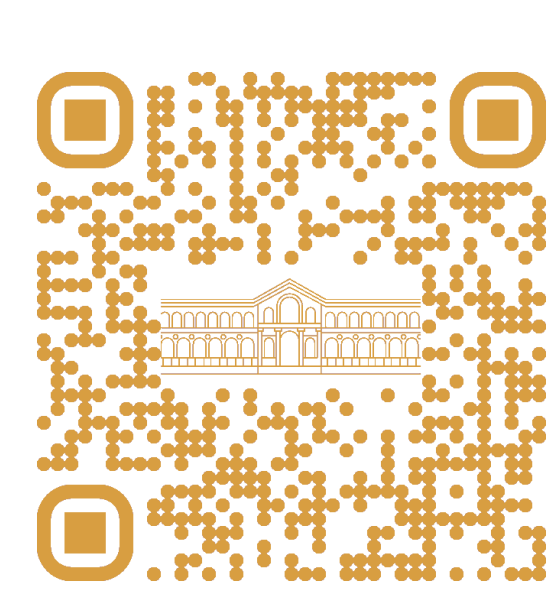




Instance-Guided Unsupervised Domain Adaptation for Robotic Semantic Segmentation



ICRA

MICHELE ANTONAZZI

micant@kth.se

LORENZO SIGNORELLI

lorenzo.signorelli@student.unimi.it

MATTEO LUPERTO

matteo.luperto@unimi.it

NICOLA BASILICO

nicola.basilico@unimi.it

Context

Domain shift problem in semantic segmentation for indoor robotics

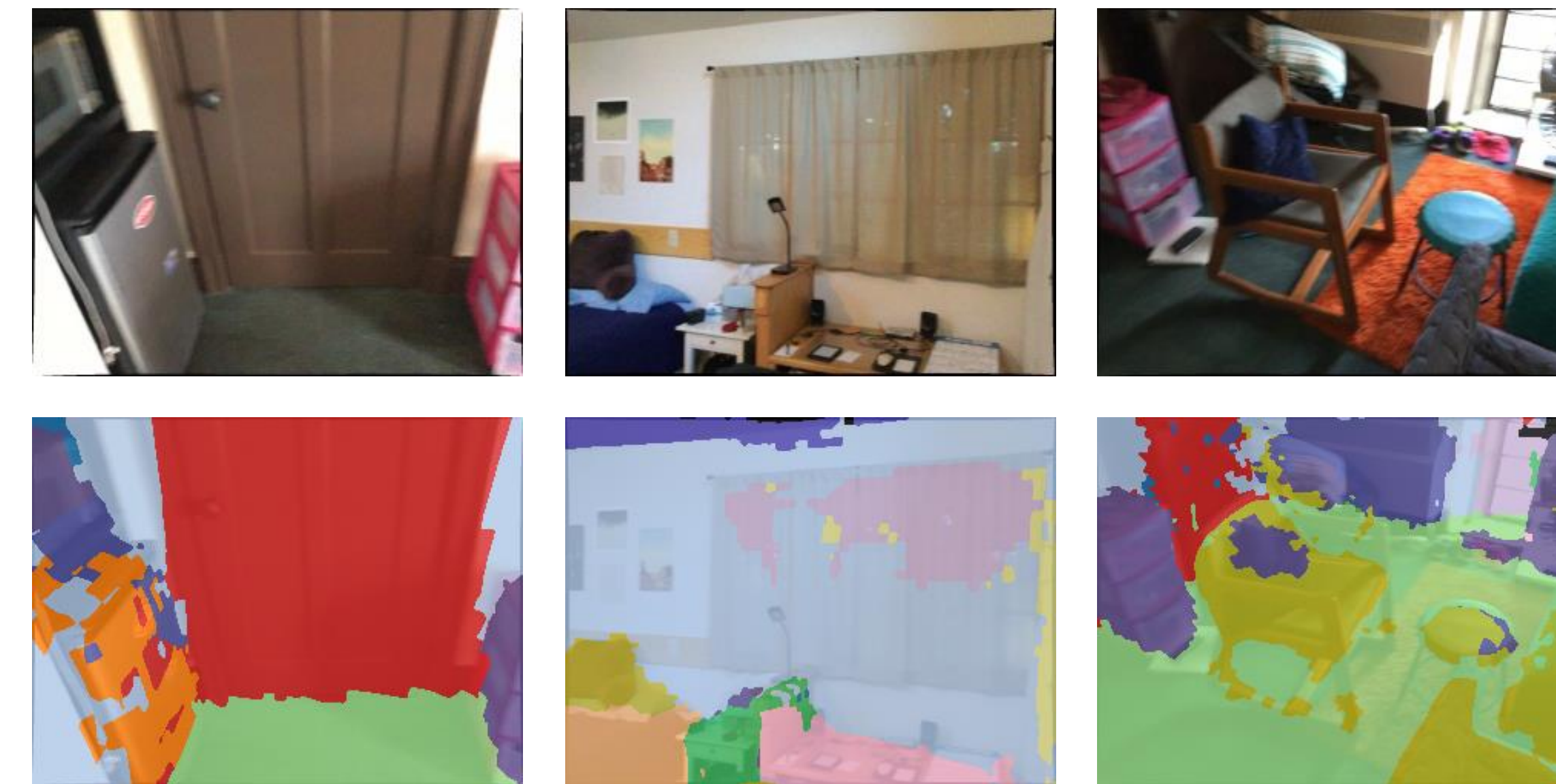
- Training on a **source domain**: general-purpose datasets, often in controlled conditions
 - Testing on a **target domain**: unseen environments, often with novel visual aspects
 - Fine-tuning with manual annotations is not practical in real-world robot deployments
 - Self-supervision with pseudo-labels is a common solution for unsupervised adaptation^[1]
- Performance degradation



Motivation

Multi-view consistent self-supervision

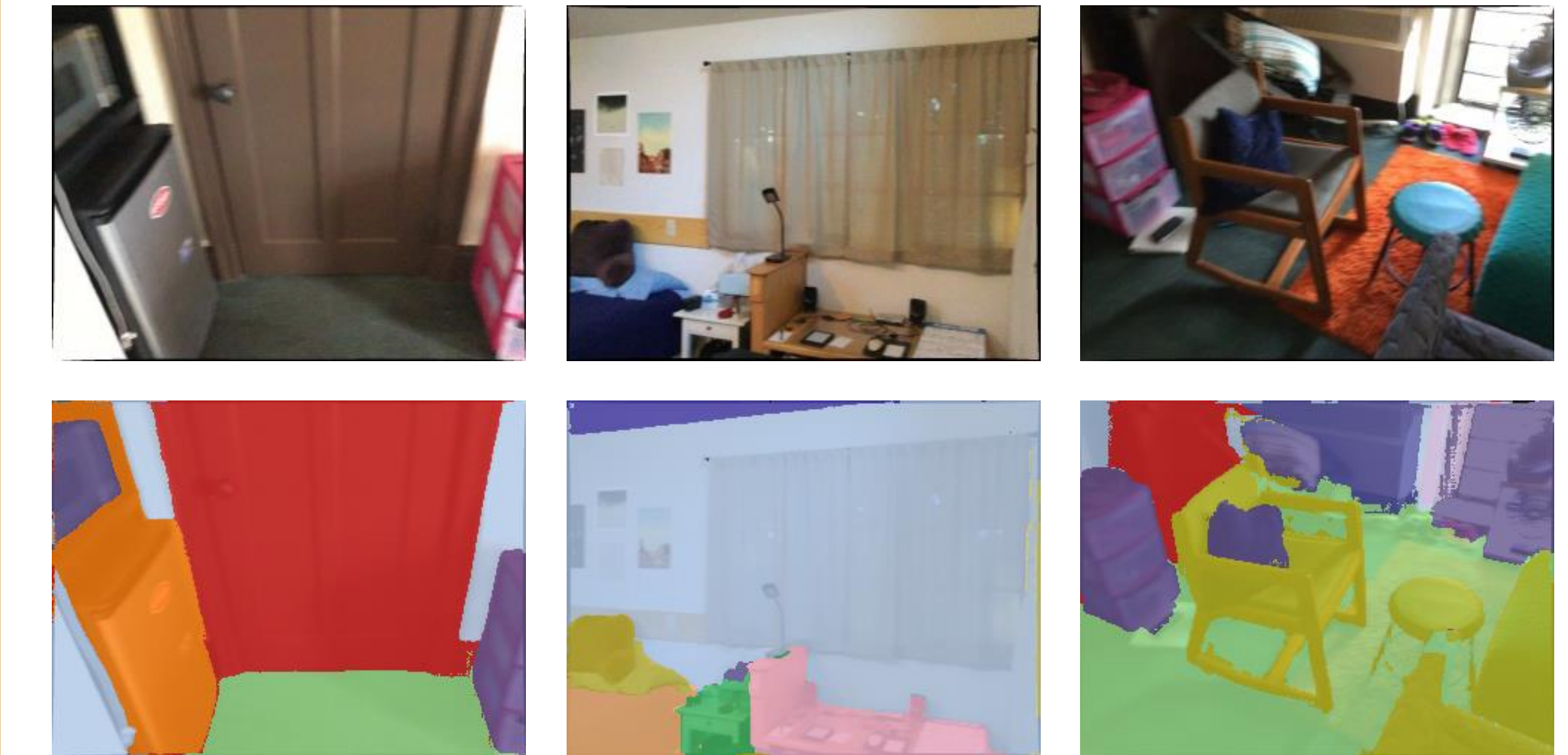
- Aggregating pseudo-labels from different point of views into the same 3D region (voxels in a 3D map) to force spatial consistency
- Obtaining spatially consistent pseudo-labels from the map
- **Limitations:**
 - Visual artifacts produced by the rendering process
 - Multiple categories can be assigned to the same object



Solution

Instance-guided unsupervised adaptation

- Revise pseudo-labels exploiting object-instance consistency: propagate pseudo-labels using guidance from an instance map
- We exploit the zero-shot instance segmentation model (Segment Anything v2^[3]) devising two complementary prompting strategies



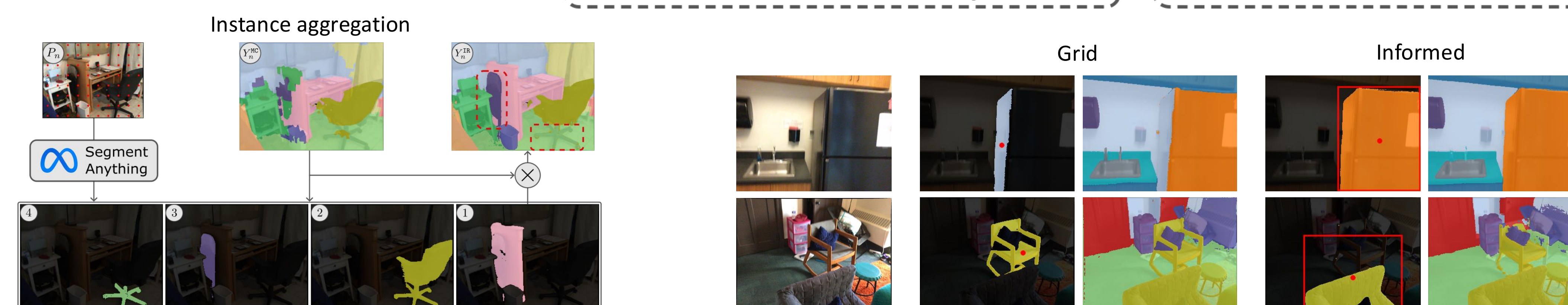
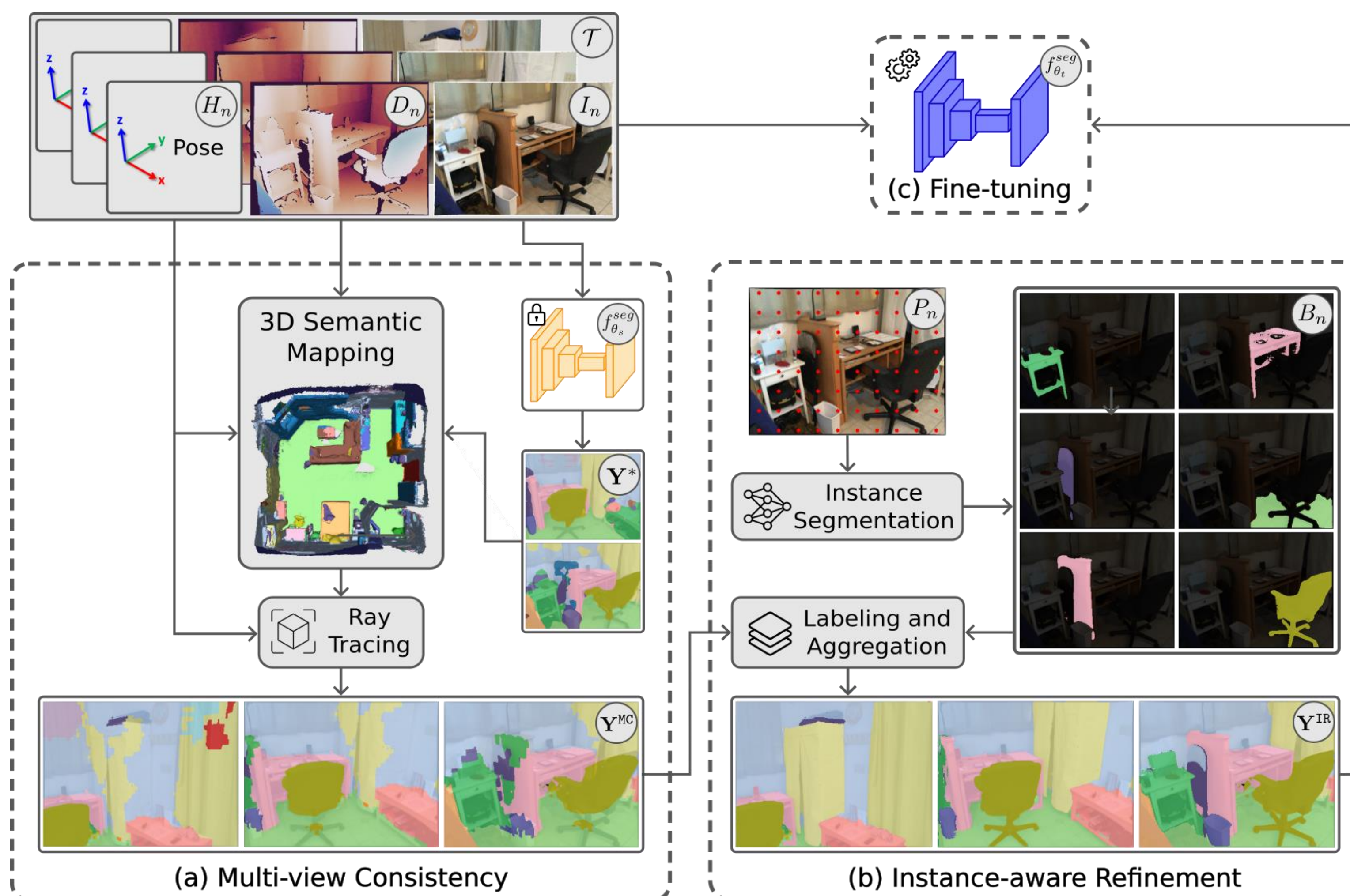
Method

a) Multi-view consistency

- Spatially-consistent pseudo-labels $\mathbf{Y}^{MC}$ are ray-traced from the map

b) Multi-view consistency

- Object instances B are obtained using SAM v2
 - **Grid**: the prompts are a grid of points uniformly distributed in the image
 - **Informed**: point and bounding box pairs derived from the pseudo-labels
 - **Combined**: pseudo-labels obtained with both strategies are used during training
- Object instances are labeled using the object categories in $\mathbf{Y}_n^{MC}$
- Labeled instances are aggregated in an instance-refined pseudo-label $\mathbf{Y}_n^{IR}$



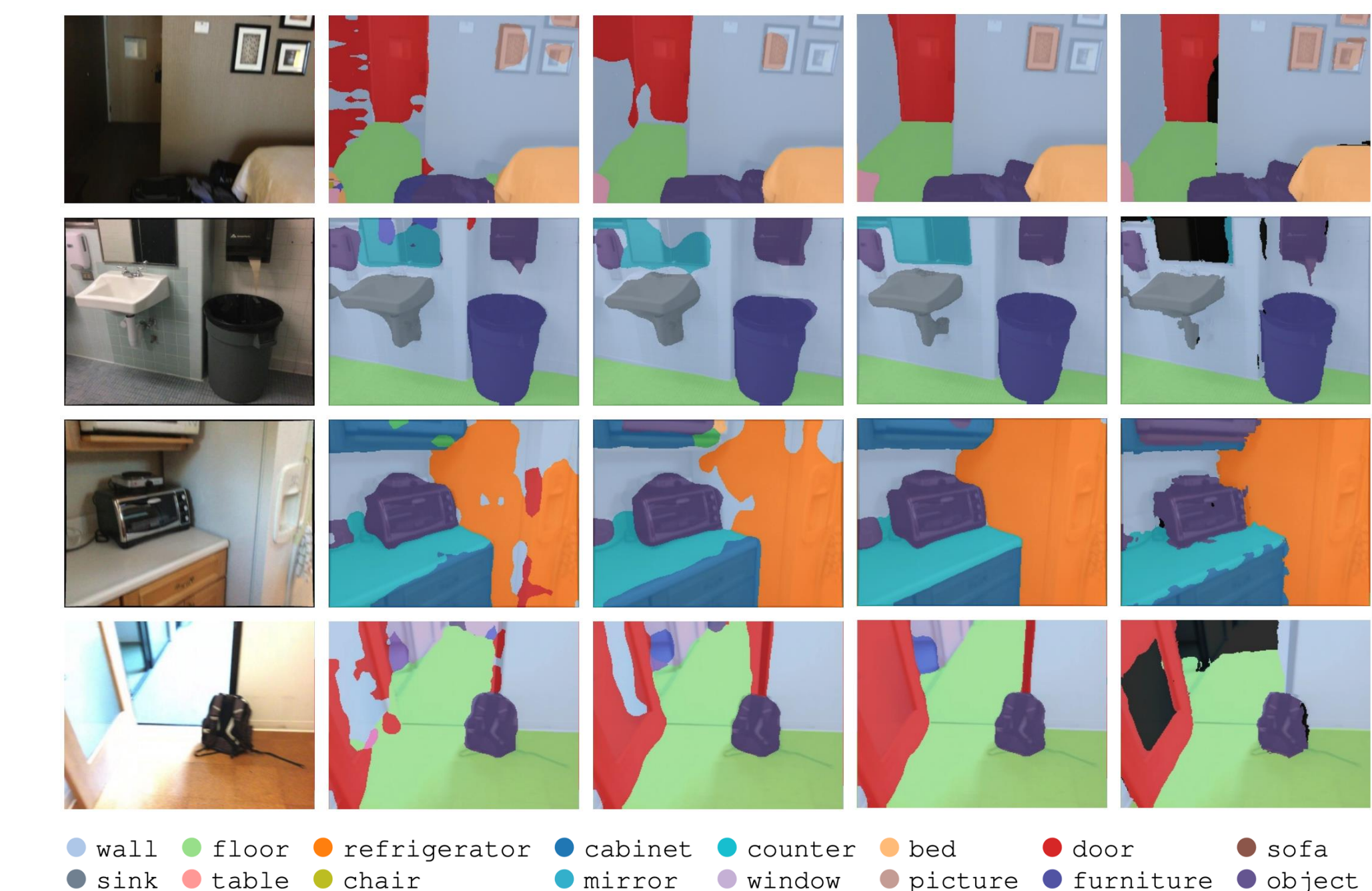
Results

Evaluation protocol

- **Dataset**: ScanNet^[4], environments 1-10 for adaptation/testing, 11-707 for initializing the source model
- **Model**: DeepLab v3
- **Mapping framework**: Kimera semantics
- **Baseline**: multi-view consistency^[2] (MC)

Results

- Our instance refinement (IR) step improves pseudo-labels quality
- Fine-tuning with our improved pseudo-labels improve model performance
- Prompting strategies performs similarly but Combined wins



Env	Source	Baseline		Our method		Our method		Our method	
		MC	Δ_{PT}	IR _G	Δ_{MC}	IR _I	Δ_{MC}	IR _{GI}	Δ_{MC}
1	47,1	49,6	5,3%	55	10,9%	51,2	3,2%	<u>54,3</u>	9,5%
2	44,1	35,2	-20,2%	39,6	12,5%	<u>39,6</u>	12,5%	39,3	11,6%
3	30,6	30,5	-0,3%	32,8	7,5%	34,4	12,8%	34,7	13,8%
4	55,3	55,8	0,9%	<u>57,2</u>	2,5%	56,2	0,7%	57,7	3,4%
6	49,2	50,5	2,6%	52,8	4,6%	54,7	8,3%	<u>53</u>	5,0%
7	52	54,4	4,6%	55,5	2%	57,2	5,1%	<u>56,2</u>	3,3%
10	57,5	63,2	9,9%	<u>68,7</u>	8,7%	67,6	7,0%	68,7	8,7%
Avg	48	48,5	0,4%	<u>51,7</u>	7%	51,6	7,1%	52,0	7,9%

References

- [1] M. Schwonberg, et al., "Survey on unsupervised domain adaptation for semantic segmentation for visual perception in automated driving," IEEE Access, 2023
- [2] J. Frey, et al., "Continual adaptation of semantic segmentation using complementary 2d-3d data representations," IEEE RAL, 2022
- [3] N. Ravi, et al., "Sam 2: Segment anything in images and videos," 2024.
- [4] A. Dai, et al., "ScanNet: Richly-annotated 3d reconstructions of indoor scenes," in Proc. of CVPR, 2017.

Introduction

Context

- We consider a fleet of robots deployed in different indoor environments that need to perform object detection
- This ability is essential to carry out high-level tasks useful in several contexts^[1]

Service Robots



Assistive Robots

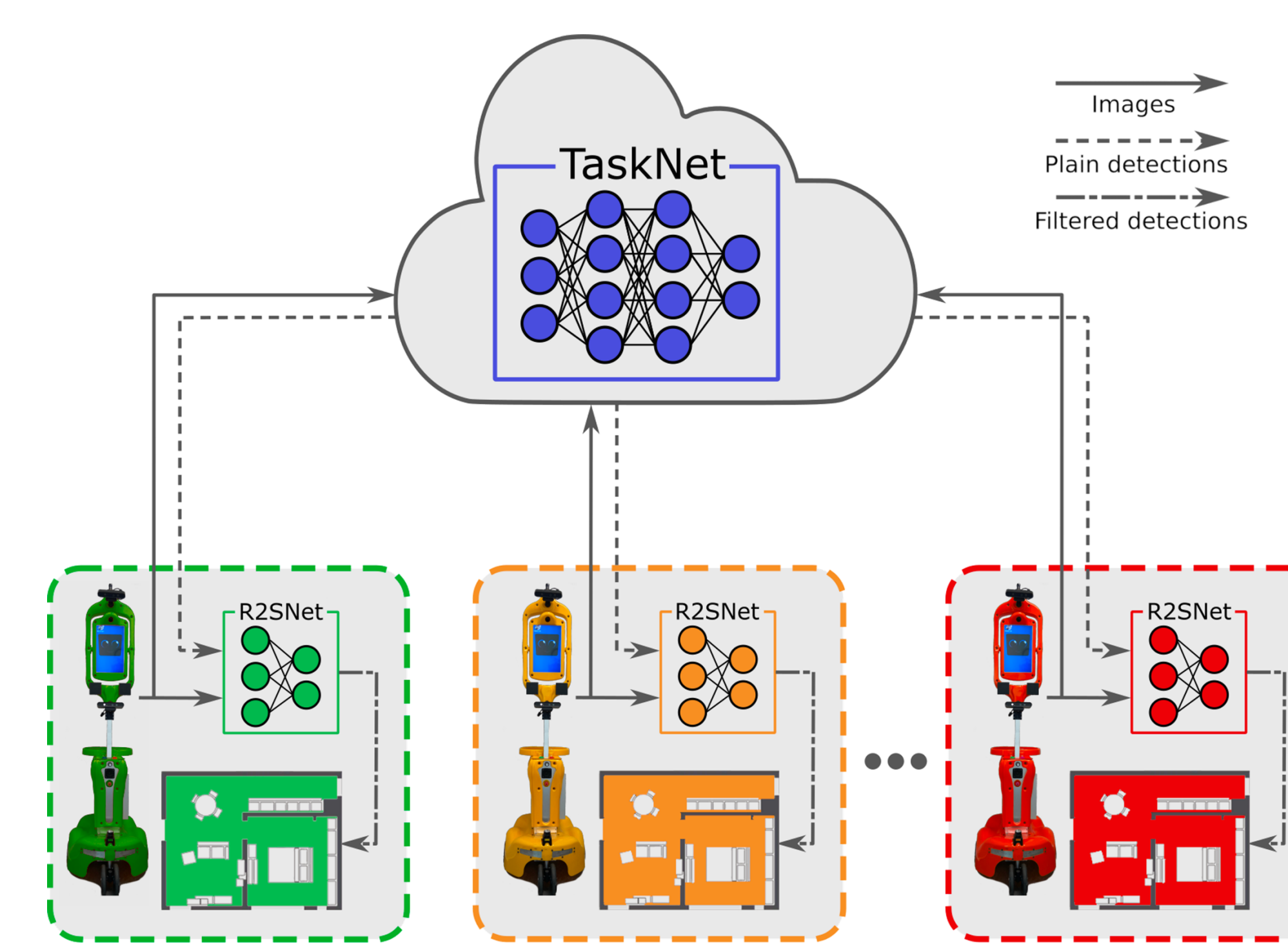


Robots as Computationally Limited Autonomous Agents

- A straightforward approach is to plug and play publicly-available Deep Neural Networks (DNNs) for object detection (OD)
- Running deep learning-based models on mobile robots is prohibitive
 - Low-powered and affordable hardware configuration
 - Limited computational capabilities affect real-time inference
 - Energy-preservation constraints for long-term autonomy

Cloud Robotics

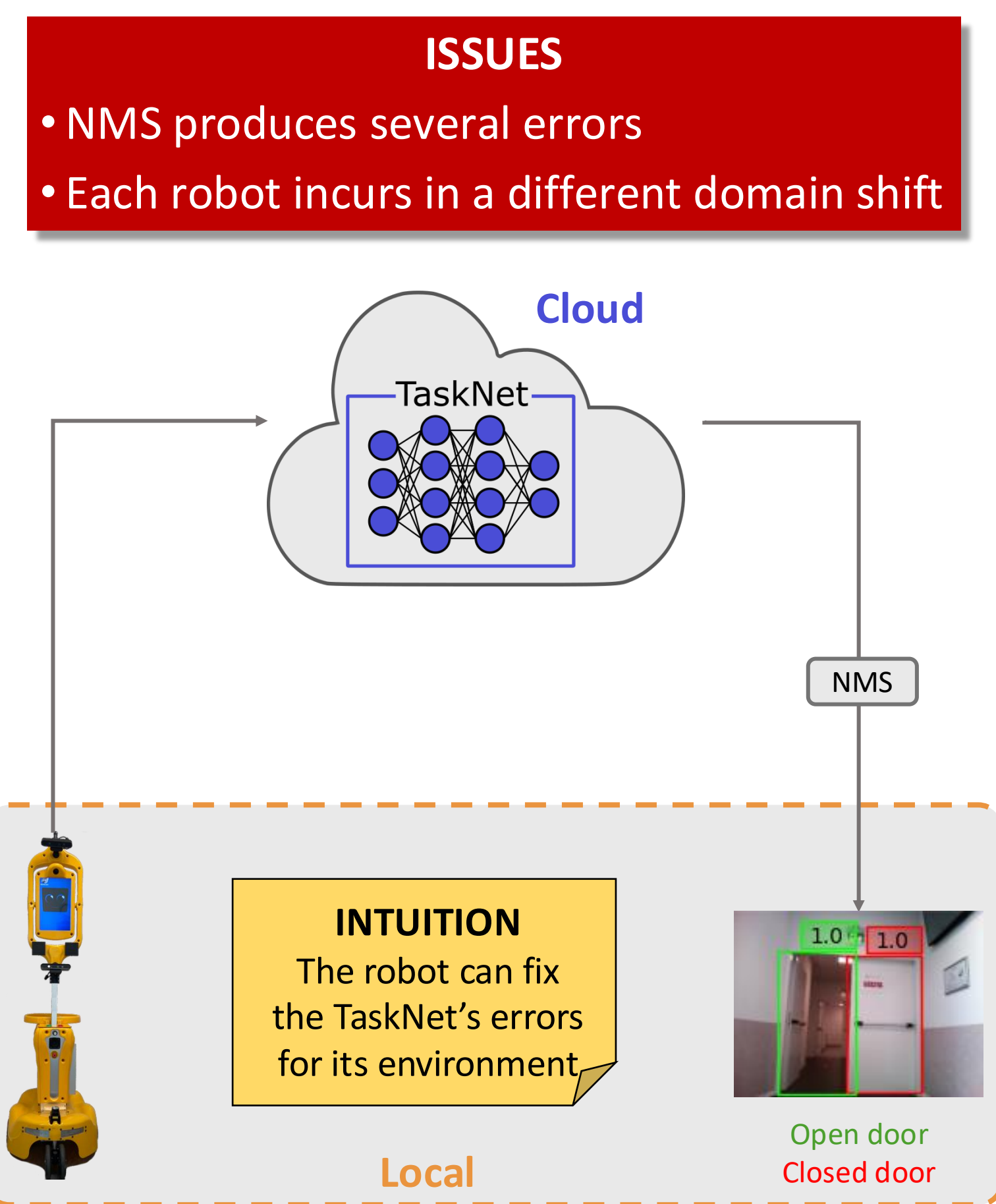
- Offloading computationally intensive inference tasks to third-party cloud services running DNNs, here called TaskNets^[2]
- Domain shift degrades the TaskNet's performance
- Classical domain adaptation^[3] cannot be applied
 - The TaskNet is inaccessible
 - Train, deploy, and maintain a TaskNet for each robot is expensive



Preliminaries

Object Detection over the Cloud

- The robot sends remotely its perceptions (RGB images)
- The TaskNet predicts a dense set of object proposals $\hat{Y} = \{\hat{y}\}$
- Bounding boxes are expressed as $\hat{y} = [\hat{c}_x, \hat{c}_y, \hat{w}, \hat{h}, \hat{c}, \text{hot}(\hat{o})]$
 - $\hat{c}_x, \hat{c}_y$ are the center coordinates
 - $\hat{w}, \hat{h}$ are width and height
 - $\hat{c}, \text{hot}(\hat{o})$ are the confidence and the one-hot encoded label
- $\hat{Y}$ is filtered using Non-Maximum Suppression (NMS)
- The remaining bounding boxes are sent back to the robot



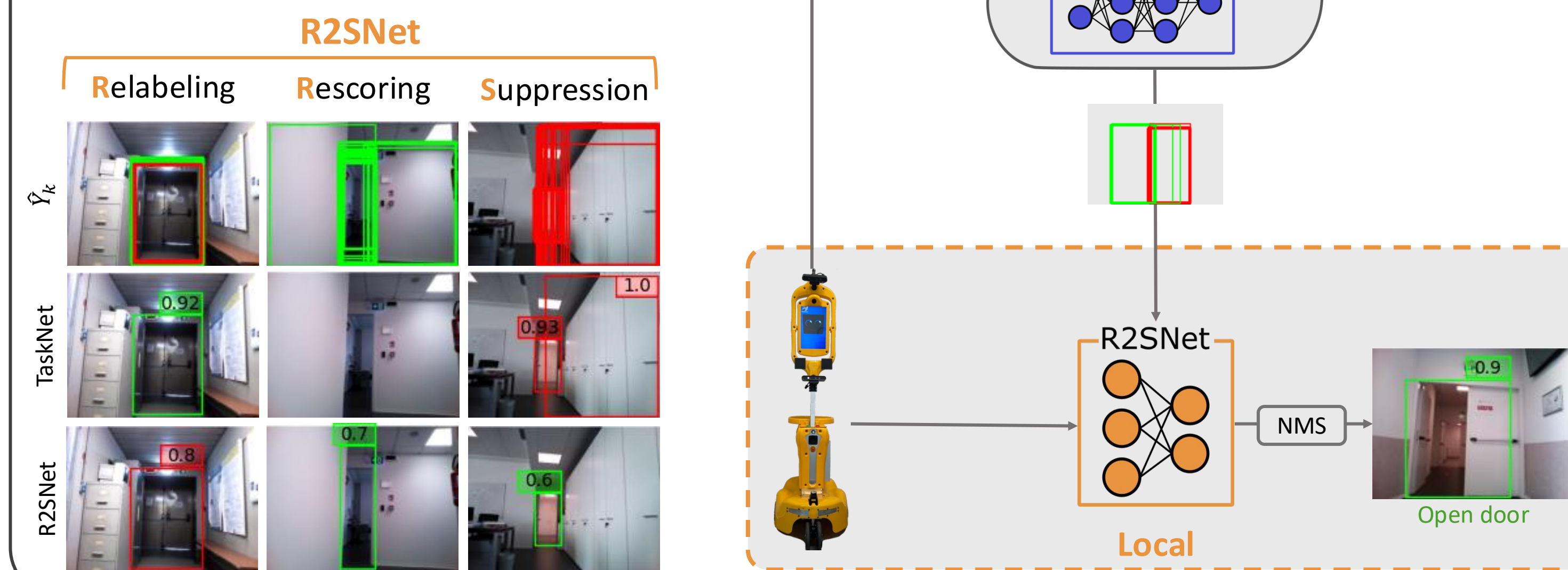
Approach

Downstream Proposal Refinement

- The robot receives $\hat{Y}$ and selects the first k most confident, obtaining $\hat{Y}_k$
- It refines their parameters with a lightweight DNN which performs 3 corrective actions
- $\hat{Y}_k$ is then filtered with NMS

BENEFITS

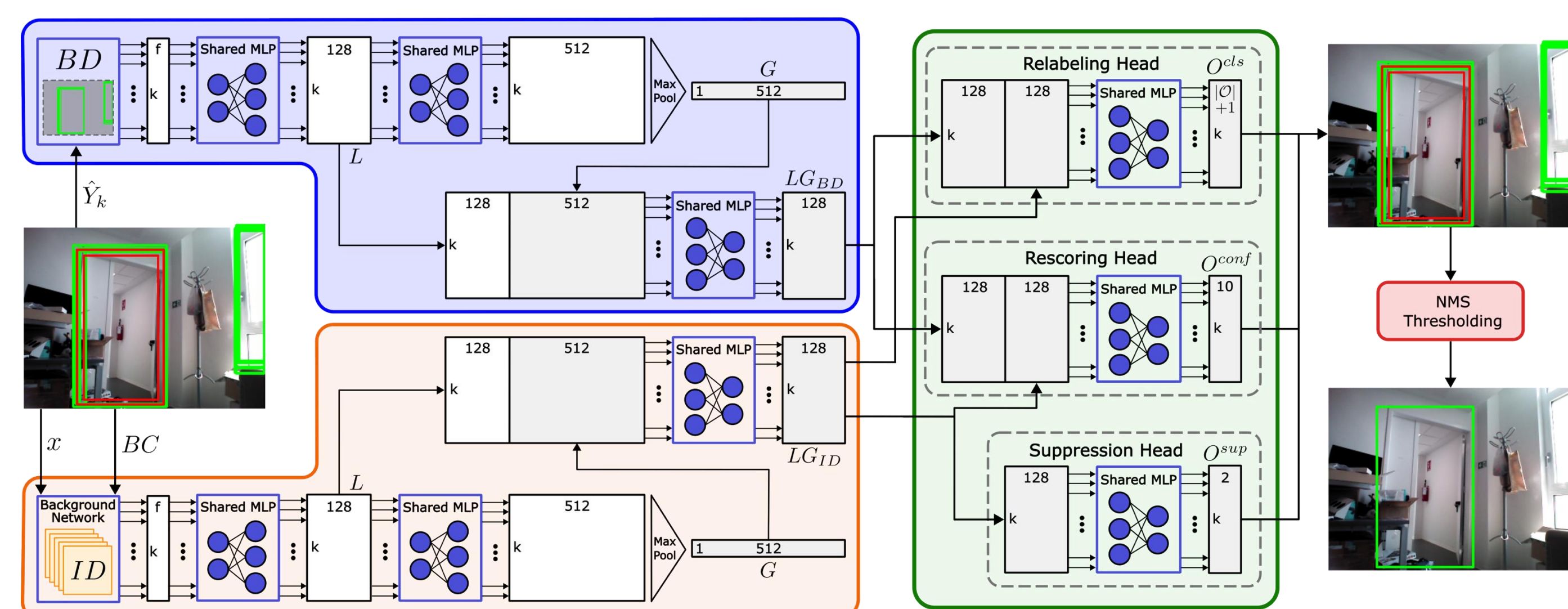
- Horizontally-scalable adaptation
- Computationally affordable by robots



Architecture

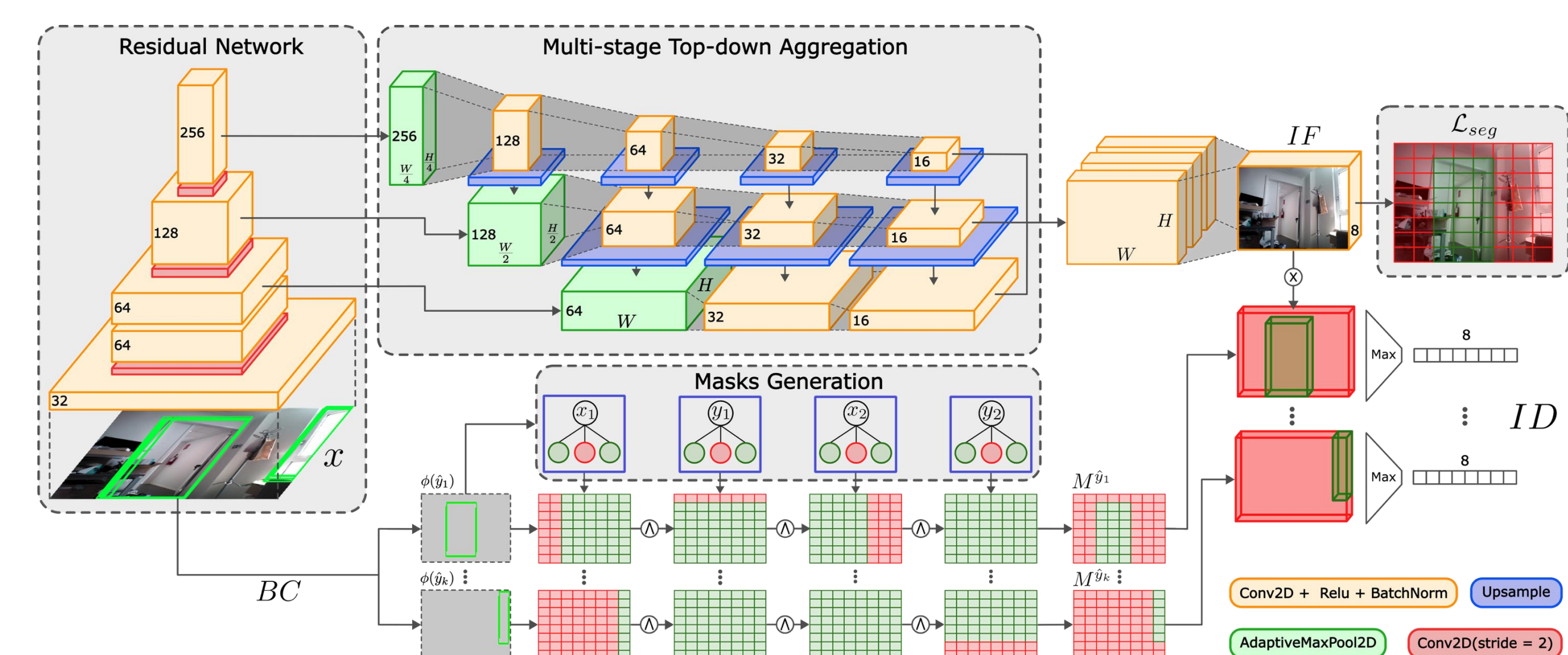
R2SNet Architecture

- Bounding boxes are expressed with two different descriptors:
 - Bounding-box Descriptors (BD): parameters of proposals received by the TaskNet
 - Image Descriptors (ID): visual features extracted by the Background Feature Network (BFNet)
- BD and ID are processed by two symmetric networks inspired by PointNet^[4]
 - Local features (L) are extracted through shared MLPs and Global features (G) with a $\max$ operator
 - Local and global features are then concatenated and mixed with shared MLPs in an embedding LG
- The mixed features are fed into 3 heads to perform relabeling, rescoring, and suppression



BFNet Architecture

- Produces an image feature map IF with dimension $[W, H, 8]$
 - Extracts a multi-scale embeddings using a residual network
 - The last 3 levels are processed by 3 parallel convolutional networks and top-down aggregated
- Produces a binary masks M for each proposal
 - 4 MLPs with fixed weights and biases
 - Each MLP extracts a partial mask for each coordinate that are aggregated with an and operator
- Masks are multiplied with IF and then maxpooled obtaining visual features for each proposals



Evaluation

- D_{DD2} : a real dataset (called DeepDoors2) with $\approx 3k$ examples
- D_G : photorealistic dataset obtained with

Train the TaskNet (Faster R-CNN)

Exp.	Mean			
	mAP $\uparrow$	TP $\uparrow$	FP $\downarrow$	BFD $\downarrow$
TaskNet	30	36%	7%	20%

Performance increases even with a few data

Ablation study			All heads contribute to domain adaptation			
Rel.	Res.	Sup.	mAP $\uparrow$	TP $\uparrow$	FP $\downarrow$	BFD $\downarrow$
✓			34	44%	10%	35%
	✓		44	48%	4%	6%
		✓	41	54%	15%	34%

